How Managers Matter

School leadership, teacher effort, and student learning under mandatory principal turnover

Paper idea for discussion. October 2026

How do school managers matter?

Whether	Does the managerial input matter once the selection of workers is shut down?
Through what	Teacher effort and practice, classroom organization, parental inputs: the production process behind principal value added.
For whom	Teacher wellbeing and family trust are welfare-relevant outcomes, not only mediators.
Who	How dispersed are manager effects, and which pre-determined traits predict a good manager.

Two stories, one design

Selection	Spain does not attract or select its best managers: no pay or career incentives, posts left vacant and filled by the administration, single candidates, committees that pass over the top-merit candidate.
Causal effect	What a change of manager does to a school, identified from mandatory exits, in a system where the selection channel is shut and the incoming manager is not selected on quality.
Link	The selection story describes who arrives and how; the causal story measures what the arrival does. The first is the interpretation of the second.

Motivating facts, Madrid 2015–2018 (correlational)

Selection	In 39% of contested competitions (51 of 132) the winner is not the top-merit candidate. Many competitions draw a single candidate or none.
Failed succession	Schools where no candidate is selected: achievement falls about 0.20 SD after the failure and stays down; climate does not move; the incoming principal reports lower satisfaction (−0.26 SD) and more regret (+0.20 SD).
Who fails	Not poorer or lower-achieving schools. Before the failure, their principals are already less satisfied and more regretful.

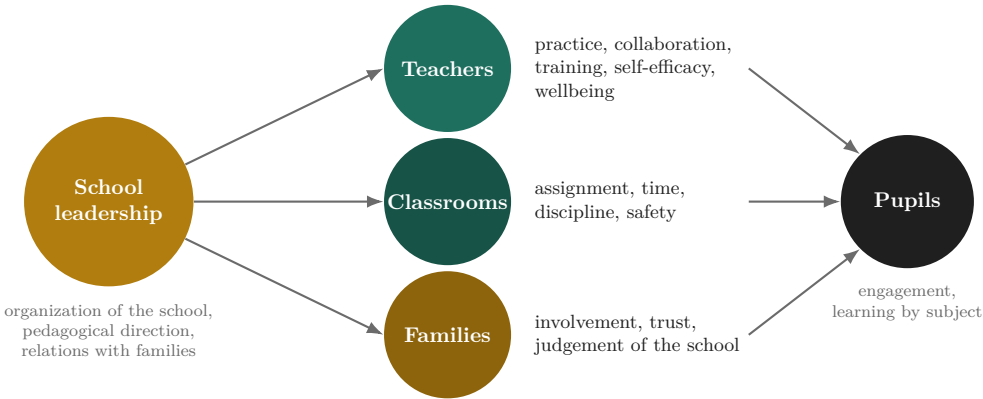
The last row is the selection problem of the turnover literature (Miller, 2013), observed directly. Mandatory exits remove it.

Literature

Value added	0.05–0.10 SD per SD of principal quality (Branch, Hanushek, and Rivkin, 2012; Böhlmark et al., 2016); 0.03 math, 0.01 reading net of school-year shocks (Bartanen, Husain, and Liebowitz, 2024).
Turnover	−0.03 SD and higher teacher exit after a change of principal (Bartanen, Grissom, and Rogers, 2019).
Mechanism	Selection and retention of teachers (Grissom and Bartanen, 2019; Bartanen and Grissom, 2023; Muñoz and Prem, 2024).
Unidentified	Effects on the practice and effort of a fixed teaching staff, on classroom organization, and on parental inputs.

Recent work

Causal map



Exogenous variation enters at the leadership node. Every other node is observed.

Setting

Managers	Principals are teachers selected for four-year terms, without authority over hiring, dismissal, pay, or curriculum.
Selection	A per-school committee (three administration, two school) scores objective merits (0–15) and a project. Outcomes: contested, single candidate, or failed and filled by the administration.
Labor input	Fixed across a change of principal. The selection channel is shut; what remains is the managerial input.
Mandatory exits	Retirement at 65 and a two-renewal limit (twelve years), both pre-determined.

Identification



Instrument	A mandatory exit in 2016, 2017, or 2018, determined by age or tenure.
Exclusion	The exit date affects outcomes only through the change of principal.
Tests	No pre-trends in any outcome. Retiring principals' schools balanced on pre-period characteristics: exits occur across the distribution, not in the worst schools.
Benchmark	Voluntary and failed successions, estimated separately: the selection on unobservables shown on the facts slide.

Reduced form

τ_s: year of the mandatory exit in school s. i: teacher, classroom, pupil, or parent.

$$Y_{ist} = \alpha_s + \lambda_t + \sum_{k \neq -1} \beta_k \, \mathbf{1}[\, t - \tau_s = k \,] + \varepsilon_{ist}$$

$\beta_{k \geq 0}$	Dynamic treatment effect k years after the exit.
$\beta_{k < 0}$	Pre-trends.
Unit	The school is the unit of treatment; i are repeated cross-sections within it. Not-yet-treated controls; staggered-robust estimator; clustered by school.

Local average treatment effect

$Z_{st} = \mathbf{1}[t \geq \tau_s]$, $D_{st} = \mathbf{1}[\text{new principal}]$.

$$D_{st} = \pi Z_{st} + \alpha_s + \lambda_t + u_{st}$$
$$Y_{ist} = \beta \widehat{D}_{st} + \alpha_s + \lambda_t + \varepsilon_{ist}$$

Compliers	At 65: a long-tenured incumbent replaced. At twelve years: a younger incumbent replaced. Two complier groups, one parameter to compare.
Decomposition	$\beta = \theta_{\text{new}} - \theta_{\text{old}}$. The incumbent's pre-exit staff rating and merits separate losing a good manager from receiving a worse one.
Panels	Same teacher (tutors re-observed) and same pupil (grade 3 to 6) where the data allow; to be verified.

Variance of manager effects; heterogeneity

$Y_{st} = \mu_s + \theta_{p(s,t)} + e_{st}$. Around a mandatory exit, θ_{old} is replaced by θ_{new}:

$$\text{Var}(\Delta Y_{st} \mid Z_{st} = 1) - \text{Var}(\Delta Y_{st} \mid Z_{st} = 0) = 2\sigma^2_\theta \qquad Y_{ist} = \ldots + \beta D_{st} + \delta\, D_{st} \times W_s + \varepsilon_{ist}$$

σ^2_θ	Dispersion of manager effects without ranking managers (Jones and Olken, 2005).
W_s	How the vacancy was filled (contested, single, failed); successor's merits and age relative to the incumbent; change in the principal's reported practices (LOMCE questionnaire); change in the school's educational project (text).
δ	Treatment effect heterogeneity by the dimensions of the change.

Extension: rules versus discretion in selecting managers

Rule	The merit score, observed for every admitted candidate.
Discretion	The committee's choice; in 39% of contested cases it overrides the merit ranking.
Instrument	Administration members are named per committee (BOCM). If they sit on several committees and are not assigned on school characteristics, their propensity to override or to fail a competition instruments who is selected (Hoffman, Kahn, and Li, 2018).
Question	Does discretion select better managers than the rule?

Outcomes and expected timing

Input	Outcome	Source	When
Classrooms	Allocation of teachers and support hours	Timetables	Year 1
Classrooms	Discipline, climate	Pupil and teacher reports	Years 1–2
Families	Involvement, trust, satisfaction	Parent survey	Years 1–2
Teachers	Self-efficacy, wellbeing, collaboration, training	Teacher survey	Years 1–2
Teachers	Classroom practice	Teacher survey	Years 2–3
Output	Engagement; achievement by subject	Pupil survey, tests	Last

Pre-registered indices by input.

Data

Schools	Universe of Madrid public primary schools, 2015–2018: principal questionnaire (age, sex, practices, satisfaction); teacher questionnaire; timetables; all pupils and parents by classroom; test scores by subject.
Selection	Official records of the 2015–2018 competitions (Consejería): schools called, admitted candidates with merit scores in four components, winner with name and corps; 581 schools; 2018 project scores for winners; 2017 exclusions with cause.
To add	Appointment and renewal history from BOCM (tenure clock); committee members; school documents from archived websites; public school results from 2022.

Open questions

- **Power.** Number of mandatory exits among primary schools, 2016–2018; minimum detectable effects by outcome at the school level.
- **Panels.** Share of tutors and pupils re-observed; whether within-unit designs are feasible.
- **Links.** Successor's name and corps to the prior-year teacher questionnaire; committee members across schools.

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Appendix: recent work on school and public-sector managers

Title	Journal	Year
Principal effects on teacher working conditions (Rahaman and Koedel)	CALDER WP	2026
Rethinking principal effects on student outcomes	J. Public Economics	2024
Managers' productivity and recruitment in the public sector: the case of school principals	AEJ: Economic Policy	2024
Personnel management and school productivity: evidence from India	Economic Journal	2024
The lasting impacts of middle school principals	NBER WP	2024
Managers in public schools (Tsao)	Princeton JMP	2024
Effects and mechanisms of CEOs quality in public education	Economic Journal	2023
School principal race, teacher racial diversity, and student achievement	J. Human Resources	2023
Managers and productivity in the public sector	Econometrica	2022

Tsao: event studies of principal moves and retirements, Kentucky and Texas; effects on scores, absenteeism, teacher retention. Rahaman and Koedel: variance of principal effects on 20 teacher-reported working conditions, Illinois.

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